Multiplication of algebraic terms

1 Simplify the following:

a $10 \times 6u$
b $7 \times (-3u)$
c $5x \times 2$
d $11 \times 3y$
e $(-8x) \times 9$
f $(-12) \times (-2u)$
g $(-3w) \times 2$
h $10 \times (-3y)$

2 Simplify the following:

a $x \times x$
b $2x \times 3x$
c $4y \times y$
d $3a \times 5a$
e $8x \times 2xy$
f $9b \times 3b$
g $(10a)^2$
h $(4x)^2$

3 Simplify the following:

a $9 \times m \times n \times 8$
b $w \times 4 \times z \times 6$
c $10 \times (-r) \times s \times (-5)$
d $9r \times 6s$
e $6u^2 \times 7v^8$
f $16p^3 \times 14q^3$
g $(-2a) \times (-4b)$
h $3w \times (-7z)$
i $(-5r) \times (-4s)$
j $\left(-10r^8\right) \times 6s^7$
k $4p^5 \times \left(-3q^5\right)$
l $2a^4 \times \left(-5b^3\right)$

4 Simplify the following:

a $6r \times 2 \times 8s$
b $7w \times 9x \times 10y$
c $(-2w) \times (-4x) \times (-10y)$
d $(-8h) \times 5k \times (-3r) \times (-4s)$
e $5x \times 2y \times (-9)$
f $4a \times (-2a) \times 10a$
g $(-5x) \times (-2x) \times (-3y) \times (-3)$
h $(-2h) \times 4k \times (-j) \times (-5i)$

Division of algebraic terms

5 Simplify the following:

a $\dfrac{2x}{2}$
b $\dfrac{15v}{5}$
c $\dfrac{5m}{20}$
d $\dfrac{n}{4n}$
e $\dfrac{12xy}{12}$
f $\dfrac{63pq}{9p}$
g $\dfrac{12mn}{15m}$
h $\dfrac{6r}{rw}$
i $\dfrac{pr}{pqr}$
j $\dfrac{10u^6v^4}{u^6}$
k $\dfrac{-24a}{4}$
l $\dfrac{-11y}{y}$
m $\dfrac{y}{-11y}$
n $\dfrac{-12u}{3u}$
o $\dfrac{-4m}{-9m}$
p $\dfrac{-abc}{b}$

q $\dfrac{k}{-jk}$

r $\dfrac{-6j}{jk}$

s $\dfrac{-ac}{abc}$

t $\dfrac{-2b^3}{3b^3}$

u $\dfrac{-3r^3w^5}{r^3}$

v $\dfrac{12n}{n}$

6 Simplify the following:

a $5m \div 40$

b $20wz \div 4wz$

c $10r^6 \div 5r^6$

d $\left(-24r^4\right) \div 6r^4$

e $(-44rs) \div 4r$

f $(-36uv) \div (-6uv)$

g $10mn \div 5m$

h $18xy \div 6y$

i $27r^2 \div 9r$

j $\left(-20x^4\right) \div 10x^4$

k $\left(-22x^2y\right) \div -2xy$

l $(-50abc) \div (-5ab)$

7 Simplify the following:

a $\dfrac{2x \times 3y}{xy}$

b $\dfrac{(-4x) \times 5y}{-10xy}$

c $\dfrac{6x \times 4xy}{8y}$

d $\dfrac{7x \times 4y}{2x \times y}$

e $\dfrac{10a \times 3b}{5b \times 2}$

f $\dfrac{12x \times 6y}{8x \times 2y}$

g $9x \times 4x \div 2x$

h $(-5x) \times 8y \div 10y$

i $11xy \times 3y \div y$

j $20y \div 4 \times 3y$

k $(10y \div 2y) \times (4x \div 2)$

l $(6x \times 3y) \div (9x \times 2)$

Applications

8 While Judy is packing rectangular boxes into crates, she notices that each crate is 12 times wider than the width of one box, and 11 times longer than the length of one box. Judy wants to know the greatest number of boxes she can pack into each crate.

Let the length of one box be L cm, and the width of one box be W cm.

a Find an expression for the volume of one box with a height of 44 cm.

b Find an expression for the volume of a crate of height H cm.

c Find an expression for the number of identical boxes that Judy can fit into each crate.

d If the crate is 88 cm high, calculate how many boxes can Judy fit into each crate.

9 Aaron have books to be placed on a shelf which is 6 times wider than the width of the book and 8 times longer than the length of the book. Aaron wants to know the greatest number of books he can place into the shelf.

Let the length of one book be x cm, and the width of one book be y cm.

a Find an expression for the volume of one book with a height of 6 cm.

b Find an expression for the volume of the shelf of thickness z cm.

c Find an expression for the number of books that Aaron can fit into the shelf.

d If the shelf is 15 cm thick, how many books can Aaron fit into the shelf?